O(n2) Time Complexcity

O(n²) means the time taken grows **proportionally to the square of input size**.
`An algorithm is O(n²) if for each element, it processes all other elements`

```python
n = 5

for i in range(n):
    for j in range(n):
        print(i, j)
```

Common O(n²) Examples in Python

1. Comparing all pairs

```python
arr = [1, 2, 3]

for i in arr:
    for j in arr:
        print(i, j)
```

2. Bubble Sort (Worst Case)

```python
arr = [5, 3, 2, 4]

for i in range(len(arr)):
    for j in range(len(arr) - 1):
        if arr[j] > arr[j+1]:
            arr[j], arr[j+1] = arr[j+1], arr[j]
```

3. Finding duplicates

```python
arr = [1, 2, 3, 1]

for i in range(len(arr)):
    for j in range(i + 1, len(arr)):
        if arr[i] == arr[j]:
            print("Duplicate found")
```

4. Triangular loop (still O(n²))

```python
for i in range(n):
    for j in range(i, n):
        pass
```